

Gradient Descent Worked Example

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Problem setup

We use a linear model with a bias and two features:

$$\hat{y} = b_0 + w_1x_1 + w_2x_2.$$

Loss (Sum of Squared Residuals, SSR) over the 4 samples:

$$L(b_0, w_1, w_2) = \sum_{i=1}^4 (y_i - \hat{y}_i)^2.$$

Gradients (partial derivatives) for SSR:

$$\frac{\partial L}{\partial b_0} = -2 \sum_{i=1}^4 (y_i - \hat{y}_i),$$

$$\frac{\partial L}{\partial w_1} = -2 \sum_{i=1}^4 x_1^{(i)} (y_i - \hat{y}_i),$$

$$\frac{\partial L}{\partial w_2} = -2 \sum_{i=1}^4 x_2^{(i)} (y_i - \hat{y}_i).$$

Gradient descent update (learning rate α):

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} L,$$

where $\theta = [b_0, w_1, w_2]^{\top}$.

Dataset

i	x_1	x_2	y
1	1	2	5
2	2	0	6
3	3	1	10
4	0	2	4

Initial parameter values and learning rate:

$$b_0^{(0)} = 0, \quad w_1^{(0)} = 0, \quad w_2^{(0)} = 0, \quad \alpha = 0.01.$$

Step 1: Predictions and SSR at initialization

With parameters all zero, $\hat{y}_i^{(0)} = 0$ for every i . Residuals $r_i^{(0)} = y_i - \hat{y}_i^{(0)} = y_i$:

$$\begin{aligned}r_1^{(0)} &= 5, \\r_2^{(0)} &= 6, \\r_3^{(0)} &= 10, \\r_4^{(0)} &= 4.\end{aligned}$$

SSR at initialization:

$$\begin{aligned}L^{(0)} &= 5^2 + 6^2 + 10^2 + 4^2 \\&= 25 + 36 + 100 + 16 \\&= 177.\end{aligned}$$

Step 2: Compute gradients at initialization

Because $\hat{y}_i^{(0)} = 0$, we compute gradients using y_i directly.

Gradient w.r.t. bias b_0

$$\begin{aligned}\left. \frac{\partial L}{\partial b_0} \right|_{(0)} &= -2 \sum_{i=1}^4 (y_i - \hat{y}_i^{(0)}) = -2 \sum_{i=1}^4 y_i \\&= -2(5 + 6 + 10 + 4) \\&= -2(25) = -50.\end{aligned}$$

Gradient w.r.t. w_1

Compute $x_1^{(i)} y_i$ for each sample:

$$\begin{aligned}x_1^{(1)} y_1 &= 1 \times 5 = 5, \\x_1^{(2)} y_2 &= 2 \times 6 = 12, \\x_1^{(3)} y_3 &= 3 \times 10 = 30, \\x_1^{(4)} y_4 &= 0 \times 4 = 0.\end{aligned}$$

Sum:

$$\sum_{i=1}^4 x_1^{(i)} y_i = 5 + 12 + 30 + 0 = 47.$$

Then

$$\left. \frac{\partial L}{\partial w_1} \right|_{(0)} = -2 \times 47 = -94.$$

Gradient w.r.t. w_2

Compute $x_2^{(i)}y_i$ for each sample:

$$x_2^{(1)}y_1 = 2 \times 5 = 10,$$

$$x_2^{(2)}y_2 = 0 \times 6 = 0,$$

$$x_2^{(3)}y_3 = 1 \times 10 = 10,$$

$$x_2^{(4)}y_4 = 2 \times 4 = 8.$$

Sum:

$$\sum_{i=1}^4 x_2^{(i)}y_i = 10 + 0 + 10 + 8 = 28.$$

Therefore

$$\left. \frac{\partial L}{\partial w_2} \right|_{(0)} = -2 \times 28 = -56.$$

Collecting the gradient vector at iteration 0:

$$\nabla L^{(0)} = \begin{bmatrix} -50 \\ -94 \\ -56 \end{bmatrix}.$$

Step 3: One GD update

Apply

$$\theta^{(1)} = \theta^{(0)} - \alpha \nabla L^{(0)}.$$

Component-wise (recalling $\theta^{(0)} = [0, 0, 0]$ and $\alpha = 0.01$):

$$b_0^{(1)} = 0 - 0.01 \times (-50) = 0 + 0.5 = 0.50,$$

$$w_1^{(1)} = 0 - 0.01 \times (-94) = 0 + 0.94 = 0.94,$$

$$w_2^{(1)} = 0 - 0.01 \times (-56) = 0 + 0.56 = 0.56.$$

So after one iteration:

$$\boxed{b_0^{(1)} = 0.50, \quad w_1^{(1)} = 0.94, \quad w_2^{(1)} = 0.56}.$$

Step 4: Predictions, residuals and SSR after the update

Compute $\hat{y}_i^{(1)} = b_0^{(1)} + w_1^{(1)}x_1^{(i)} + w_2^{(1)}x_2^{(i)}$ and residuals $r_i^{(1)} = y_i - \hat{y}_i^{(1)}$.

Sample 1: $(x_1, x_2, y) = (1, 2, 5)$

$$\begin{aligned}\hat{y}_1^{(1)} &= 0.50 + 0.94 \times 1 + 0.56 \times 2 \\ &= 0.50 + 0.94 + 1.12 \\ &= 2.56.\end{aligned}$$

Residual:

$$r_1^{(1)} = 5 - 2.56 = 2.44.$$

Sample 2: $(x_1, x_2, y) = (2, 0, 6)$

$$\begin{aligned}\hat{y}_2^{(1)} &= 0.50 + 0.94 \times 2 + 0.56 \times 0 \\ &= 0.50 + 1.88 + 0 \\ &= 2.38.\end{aligned}$$

Residual:

$$r_2^{(1)} = 6 - 2.38 = 3.62.$$

Sample 3: $(x_1, x_2, y) = (3, 1, 10)$

$$\begin{aligned}\hat{y}_3^{(1)} &= 0.50 + 0.94 \times 3 + 0.56 \times 1 \\ &= 0.50 + 2.82 + 0.56 \\ &= 3.88.\end{aligned}$$

Residual:

$$r_3^{(1)} = 10 - 3.88 = 6.12.$$

Sample 4: $(x_1, x_2, y) = (0, 2, 4)$

$$\begin{aligned}\hat{y}_4^{(1)} &= 0.50 + 0.94 \times 0 + 0.56 \times 2 \\ &= 0.50 + 0 + 1.12 \\ &= 1.62.\end{aligned}$$

Residual:

$$r_4^{(1)} = 4 - 1.62 = 2.38.$$

Compute SSR after update

Square each residual (showing full multiplication):

$$\begin{aligned}(r_1^{(1)})^2 &= (2.44)^2 = 2.44 \times 2.44 = 5.9536, \\ (r_2^{(1)})^2 &= (3.62)^2 = 3.62 \times 3.62 = 13.1044, \\ (r_3^{(1)})^2 &= (6.12)^2 = 6.12 \times 6.12 = 37.4544, \\ (r_4^{(1)})^2 &= (2.38)^2 = 2.38 \times 2.38 = 5.6644.\end{aligned}$$

Sum them to get $L^{(1)}$:

$$\begin{aligned} L^{(1)} &= 5.9536 + 13.1044 + 37.4544 + 5.6644 \\ &= (5.9536 + 13.1044) + (37.4544 + 5.6644) \\ &= 19.0580 + 43.1188 \\ &= 62.1768. \end{aligned}$$

Summary of results (iteration 0 $\rightarrow$ 1)

Initial parameters: $b_0^{(0)} = 0$, $w_1^{(0)} = 0$, $w_2^{(0)} = 0$,

Initial SSR: $L^{(0)} = 177$.

$$\text{Gradient at } (0) : \nabla L^{(0)} = \begin{bmatrix} -50 \\ -94 \\ -56 \end{bmatrix}.$$

Learning rate: $\alpha = 0.01$.

Parameters after 1 update: $b_0^{(1)} = 0.50$, $w_1^{(1)} = 0.94$, $w_2^{(1)} = 0.56$,

SSR after 1 update: $L^{(1)} = 62.1768$.

Interpretation

- The gradients were negative because all initial residuals were positive (predictions were 0 and actuals positive). The update therefore increased all parameters from 0 to positive numbers.
- The loss decreased substantially from 177 to 62.1768 after a single gradient-descent step with $\alpha = 0.01$, indicating a large initial improvement.
- Note that this is a single (very explicit) iteration. Convergence behavior over many iterations depends on the learning rate α and the curvature of the loss surface.